UNIQUENESS OF SIBLING DERIVED FUNCTORS IN THE PRESENCE OF ENOUGH NEGATIVE OBJECTS

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Editorial note. The earlier dump stopped before the final naturality cases. Original equiv.tex2422–2457 is retained; the prior complete recheck and this focused reading include those cases.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Triangulated categories, bounded derived categories of abelian categories, Yoneda representations of Ext, negative-Ext vanishing, TR3 and uniqueness of a compatible third arrow. Both local good-map constructions are retained.

Source reference. Original lemma 05RW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7425–7425. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06XS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8861–8861. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 06XP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8858–8858. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FWZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19864–19864. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Proposition 0.1. [2, Proposition 2.16]*; the fact that we do not need to assume vanishing of* $\mathrm{Ext}^q(N, X)$ *for* $q > 0$ *in the definition of negative objects above is due to* [1]*. Let* F *and* F' *be siblings as in Definition 0.2. Assume that* F *is fully faithful and that* $\mathcal{A}$ *has enough negative objects (see above). Then* F *and* F' *are isomorphic functors.*

Definition 0.2. Let $\mathcal{A}$ be an abelian category. Let $\mathcal{D}$ be a triangulated category. We say two exact functors of triangulated categories

$$F, F' : D^b(\mathcal{A}) \longrightarrow \mathcal{D}$$

are *siblings*, or we say F' is a *sibling* of F, if the following two conditions are satisfied

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(1) the functors $F \circ i$ and $F' \circ i$ are isomorphic where $i : \mathcal{A} \to D^b(\mathcal{A})$ is the inclusion functor, and
(2) $F(K) \cong F'(K)$ for any K in $D^b(\mathcal{A})$.

Let us make a nonstandard definition. Let $\mathcal{A}$ be an abelian category. Let us say $\mathcal{A}$ *has enough negative objects* if given any $X \in \mathrm{Ob}(\mathcal{A})$ there exists an object N such that

(1) there is a surjection $N \to X$ and
(2) $\mathrm{Hom}(X, N) = 0$.

Let us prove a couple of lemmas about this notion in order to help with the proof of Proposition 0.1.

Lemma 0.3. *Let $\mathcal{A}$ be an abelian category with enough negative objects. Let $X \in D^b(\mathcal{A})$. Let $b \in \mathbf{Z}$ with $H^i(X) = 0$ for $i > b$. Then there exists a map $N[-b] \to X$ such that the induced map $N \to H^b(X)$ is surjective and $\mathrm{Hom}(H^b(X), N) = 0$.*

Proof. Using the truncation functors we can represent X by a complex $A^a \to A^{a+1} \to \ldots \to A^b$ of objects of $\mathcal{A}$. Choose N in $\mathcal{A}$ such that there exists a surjection $t : N \to A^b$ and such that $\mathrm{Hom}(A^b, N) = 0$. Then the surjection t defines a map $N[-b] \to X$ as desired. □

Lemma 0.4. *Let $\mathcal{A}$ be an abelian category with enough negative objects. Let $f : X \to X'$ be a morphism of $D^b(\mathcal{A})$. Let $b \in \mathbf{Z}$ such that $H^i(X) = 0$ for $i > b$ and $H^i(X') = 0$ for $i \geq b$. Then there exists a map $N[-b] \to X$ such that the induced map $N \to H^b(X)$ is surjective, such that $\mathrm{Hom}(H^b(X), N) = 0$, and such that the composition $N[-b] \to X \to X'$ is zero.*

Proof. We can represent f by a map $f^\bullet : A^\bullet \to B^\bullet$ of bounded complexes of objects of $\mathcal{A}$, see for example Derived Categories, Lemma 05RW. Consider the object

$$C = \mathrm{Ker}(A^b \to A^{b+1}) \times_{\mathrm{Ker}(B^b \to B^{b+1})} B^{b-1}$$

of $\mathcal{A}$. Since $H^b(B^\bullet) = 0$ we see that $C \to H^b(A^\bullet)$ is surjective. On the other hand, the map $C \to A^b \to B^b$ is the same as the map $C \to B^{b-1} \to B^b$ and hence the composition $C[-b] \to X \to X'$ is zero. Since $\mathcal{A}$ has enough negative objects, we can find an object N which has a surjection $N \to C \oplus H^b(X)$ such that $\mathrm{Hom}(C \oplus H^b(X), N) = 0$. Then N together with the map $N[-b] \to X$ is a solution to the problem posed by the lemma. □

Proof. By part (2) of Definition 0.2 the image of the functor F' is contained in the essential image of the functor F. Hence the functor $H = F^{-1} \circ F'$ is a sibling of the identity functor. This reduces us to the case described in the next paragraph.

Let $\mathcal{D} = D^b(\mathcal{A})$. We have to show a sibling $F : \mathcal{D} \to \mathcal{D}$ of the identity functor is isomorphic to the identity functor. Given an object X of $\mathcal{D}$ let us say X has *width* $w = w(X)$ if $w \geq 0$ is minimal such that there exists an integer $a \in \mathbf{Z}$ with $H^i(X) = 0$ for $i \notin [a, a + w - 1]$. Since F is a sibling of the identity and since $F \circ [n] = [n] \circ F$ we are already given isomorphisms

$$c_X : X \to F(X)$$

for $w(X) \leq 1$ compatible with shifts. Moreover, if $X = A[-a]$ and $X' = A'[-a]$ for some $A, A' \in \mathrm{Ob}(\mathcal{A})$ then for any morphism $f : X \to X'$ the diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & X' \\ {\scriptstyle c_X}\downarrow & & \downarrow{\scriptstyle c_{X'}} \\ F(X) & \xrightarrow{F(f)} & F(X') \end{array} \tag{0.4.1}$$

is commutative.

Next, let us show that for any morphism $f : X \to X'$ with $w(X), w(X') \leq 1$ the diagram (0.4.1) commutes. If X or X' is zero, this is clear. If not then we can write $X = A[-a]$ and $X' = A'[-a']$ for unique A, A' in $\mathcal{A}$ and $a, a' \in \mathbf{Z}$. The case $a = a'$ was discussed above. If $a' > a$, then $f = 0$ (Derived Categories, Lemma 06XS) and the result is clear. If $a' < a$ then f corresponds to an element $\xi \in \mathrm{Ext}^q(A, A')$ with $q = a - a'$. Using Yoneda extensions, see Derived Categories, Section 06XP, we can find $A = A_0, A_1, \ldots, A_{q-1}, A_q = A' \in \mathrm{Ob}(\mathcal{A})$ and elements

$$\xi_i \in \mathrm{Ext}^1(A_{i-1}, A_i)$$

such that ξ is the composition $\xi_q \circ \ldots \circ \xi_1$. In other words, setting $X_i = A_i[-a+i]$ we obtain morphisms

$$X = X_0 \xrightarrow{f_1} X_1 \to \ldots \to X_{q-1} \xrightarrow{f_q} X_q = X'$$

whose compostion is f. Since the commutativity of (0.4.1) for $f_1, \ldots, f_q$ implies it for f, this reduces us to the case $q = 1$. In this case after shifting we may assume we have a distinguished triangle

$$A' \to E \to A \xrightarrow{f} A'[1]$$

Observe that E is an object of $\mathcal{A}$. Consider the following diagram

$$\begin{array}{ccccccc}
E & \longrightarrow & A & \xrightarrow{f} & A'[1] & \longrightarrow & E[1] \\
\downarrow{\scriptstyle c_E} & & \downarrow{\scriptstyle c_A} & \swarrow{\scriptstyle \epsilon} & \downarrow{\scriptstyle \gamma}{\scriptstyle c_{A'}[1]} & & \downarrow{\scriptstyle c_E[1]} \\
F(E) & \longrightarrow & F(A) & \xrightarrow{F(f)} & F(A')[1] & \longrightarrow & F(E)[1]
\end{array}$$

whose rows are distinguished triangles. The square on the right commutes already but we don't yet know that the middle square does. By the axioms of a triangulated category we can find a morphism γ which does make the diagram commute. Then $\gamma - c_{A'}[1]$ composed with $F(A')[1] \to F(E)[1]$ is zero hence we can find $\epsilon : A'[1] \to F(A)$ such that $\gamma - c_{A'}[1] = F(f) \circ \epsilon$. However, any arrow $A'[1] \to F(A)$ is zero as it is a negative ext class between objects of $\mathcal{A}$. Hence $\gamma = c_{A'}[1]$ and we conclude the middle square commutes too which is what we wanted to show.

To finish the proof we are going to argue by induction on w that there exist isomorphisms $c_X : X \to F(X)$ for all X with $w(X) \leq w$ compatible with all morphisms between such objects. The base case $w = 1$ was shown above. Assume we know the result for some $w \geq 1$.

Let X be an object with $w(X) = w+1$. Pick $a \in \mathbf{Z}$ with $H^i(X) = 0$ for $i \notin [a, a+w]$. Set $b = a + w$ so that $H^b(X)$ is nonzero. Choose $N[-b] \to X$ as in Lemma 0.3. Choose a distinguished diagram

$$N[-b] \to X \to Y \to N[-b+1]$$

Computing the long exact cohomology sequence we find $w(Y) \leq w$. Hence by induction we find the solid arrows in the following diagram

$$\begin{array}{ccccccc}
N[-b] & \longrightarrow & X & \longrightarrow & Y & \longrightarrow & N[-b+1] \\
\downarrow{\scriptstyle c_N[-b]} & & \downarrow{\scriptstyle c_{N[-b] \to X}} & & \downarrow{\scriptstyle c_Y} & & \downarrow{\scriptstyle c_N[-b+1]} \\
F(N)[-b] & \longrightarrow & F(X) & \longrightarrow & F(Y) & \longrightarrow & F(N)[-b+1]
\end{array}$$

We obtain the dotted arrow $c_{N[-b] \to X}$. By Derived Categories, Lemma 0FWZ the dotted arrow is unique because $\mathrm{Hom}(X, F(N)[-b]) \cong \mathrm{Hom}(X, N[-b]) = 0$ by our

choice of N. In fact, $c_{N[-b]\to X}$ is the unique dotted arrow making the square with vertices $X, Y, F(X), F(Y)$ commute.

Let $N'[-b] \to X$ be another map as in Lemma 0.3 and let us prove that $c_{N[-b]\to X} = c_{N'[-b]\to X}$. Observe that the map $(N \oplus N')[-b] \to X$ also satisfies the conditions of Lemma 0.3. Thus we may assume $N'[-b] \to X$ factors as $N'[-b] \to N[-b] \to X$ for some morphism $N' \to N$. Choose distinguished triangles $N[-b] \to X \to Y \to N[-b+1]$ and $N'[-b] \to X \to Y' \to N'[-b+1]$. By axiom TR3 we can find a morphism $g : Y' \to Y$ which joint with id_X and $N' \to N$ forms a morphism of triangles. Since we have (0.4.1) for g we conclude that

$$(F(X) \to F(Y)) \circ c_{N'[-b]\to X} = (F(X) \to F(Y)) \circ c_{N[-b]\to X}$$

The uniqueness of $c_{N[-b]\to X}$ pointed out in the construction above now shows that $c_{N'[-b]\to X} = c_{N[-b]\to X}$.

Thus we can now define for X of width $w+1$ the isomorphism $c_X : X \to F(X)$ as the common value of the maps $c_{N[-b]\to X}$ where $N[-b] \to X$ is as in Lemma 0.3. To finish the proof, we have to show that the diagrams (0.4.1) commute for all morphisms $f : X \to X'$ between objects with $w(X) \leq w+1$ and $w(X') \leq w+1$. Choose $a \leq b \leq a+w$ such that $H^i(X) = 0$ for $i \notin [a, b]$ and $a' \leq b' \leq a'+w$ such that $H^i(X') = 0$ for $i \notin [a', b']$. We will use induction on $(b'-a')+(b-a)$ to show the claim. (The base case is when this number is zero which is OK because $w \geq 1$.) We distinguish two cases.

Case I: $b' < b$. In this case, by Lemma 0.4 we may choose $N[-b] \to X$ as in Lemma 0.3 such that the composition $N[-b] \to X \to X'$ is zero. Choose a distuiguished triangle $N[-b] \to X \to Y \to N[-b+1]$. Since $N[-b] \to X'$ is zero, we find that f factors as $X \to Y \to X'$. Since $H^i(Y)$ is nonzero only for $i \in [a, b-1]$ we see by induction that (0.4.1) commutes for $Y \to X'$. The diagram (0.4.1) commutes for $X \to Y$ by construction if $w(X) = w+1$ and by our first induction hypothesis if $w(X) \leq w$. Hence (0.4.1) commutes for f.

Case II: $b' \geq b$. In this case we choose $N'[-b'] \to X'$ as in Lemma 0.3. We may also assume that $\mathrm{Hom}(H^{b'}(X), N') = 0$ (this is relevant only if $b' = b$), for example because we can replace N' by an object N'' which surjects onto $N' \oplus H^{b'}(X)$ and such that $\mathrm{Hom}(N' \oplus H^{b'}(X), N'') = 0$. We choose a distinguished triangle $N'[-b'] \to X' \to Y' \to N'[-b'+1]$. Since $\mathrm{Hom}(X, X') \to \mathrm{Hom}(X, Y')$ is injective by our choice of N' (details omitted) the same is true for $\mathrm{Hom}(X, F(X')) \to \mathrm{Hom}(X, F(Y'))$. Hence it suffices in this case to check that (0.4.1) commutes for the composition $X \to Y'$ of the morphisms $X \to X' \to Y'$. Since $H^i(Y')$ is nonzero only for $i \in [a', b'-1]$ we conclude by induction hypothesis. □

References

[1] Alberto Canonaco and Paolo Stellari. Fourier-Mukai functors in the supported case. *Compos. Math.*, 150(8):1349–1383, 2014.

[2] D. O. Orlov. Equivalences of derived categories and $K3$ surfaces. *J. Math. Sci. (New York)*, 84(5):1361–1381, 1997. Algebraic geometry, 7.